

Homework 6

Due Wednesday, November 1

Please use, for both problems in this homework, the nonrelativistic version of the particle's Lagrangian, that is,

$$L = \frac{1}{2}mv^2 + q\mathbf{v} \cdot \mathbf{A}(\mathbf{r}) - V(r),$$

written here in SI units. $V(r)$ is the central potential, which is nonzero for Prob. 2.

Problems:

1. Problem 23 to Chap. 9 of the textbook.
2. Problem 6 to Chap. 10 of the textbook.